

Ragavendra Raja J

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Computer Science Engineering Core

Vellore, India

2021- present

- Cumulative GPA : **8.77/10.0**

Relevant Coursework: Data Structures and Algorithms, OOPs, Operating Systems, Database Systems, Computer Networks, Data Science, Web Mining, Machine Learning, Software Engineering, Cloud Computing, Cryptography & Networks Security

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, Matlab

Frameworks: KNIME, Tableau, Apache Maven, Node.js, MongoDB, Next.js, TailwindCSS

Developer Tools: VSCode, Git, Github, Docker, Jenkins

Libraries: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, NLTK, Tkinter

PROJECTS

Optimizing Data Security in Transmission Through Lightweight Cryptography

Ongoing

- Designed an advanced cryptographic model integrating **ChaCha20** encryption with **Ikeda chaotic maps** to enhance data security in IoT environments.
- Implemented tile-based image encryption, achieving **99.6%** NPCR and passing NIST randomness tests for robustness.
- Deployed the system on GitHub Pages, improving accessibility and enhancing internal collaboration.

Advanced Hotel Recommendation System with Web Scraping and Sentiment Analysis

- Engineered an AI-powered hotel recommendation system by leveraging advanced web scraping techniques to extract and process data from online sources.
- Applied sentiment analysis to over 5,000 user reviews from **Booking.com**, utilizing NLP techniques to generate data-driven insights and enhance recommendation accuracy.
- Designed a robust data pipeline utilizing **BeautifulSoup** and **NLTK** for sentiment-driven ranking of customer feedback.

JustBroom – Hostel Room Cleaning Application Using Flask, MongoDB

- Developed a Flask-based room-cleaning application for hostel management. Designed a RESTful API using Flask to handle room cleaning requests and status updates.
- Integrated user authentication and role-based access control for secure request handling. Built a responsive front-end using HTML, CSS, and JavaScript for an improved user experience.
- Implemented MongoDB for scalable and efficient data storage.

EXTRACURRICULAR AND EXPERIENCE

Senior Core Committee Member | *DigitSquad VIT*

Jan. 2022 – Jan. 2024

- Planned, organized, and executed 5+ technical events and workshops, increasing member engagement and enhancing hands-on technical skill development.

Student Manager | *graVITas'24 (Technical Fest) & Riviera'25 (Cultural Fest)*

Sep. 2024 – Feb. 2025

- Managed sales operations for Riviera'24 & Riviera'25, one of India's top 5 cultural fests, overseeing sponsorship acquisition, event ticket sales, and merchandise distribution.

Core Member | *Youth Red Cross VIT*

Mar. 2022 – Present

- Coordinated 2 successful donation drives, mobilizing over 300 community members to contribute essential goods and services, resulting in a significant boost in local engagement and measurable social impact.

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals

Software Engineering Job Simulation Goldman Sachs

Game Theory Stanford University

SQL Injection Attacks EC Council

Problem Solving (Intermediate) HackerRank